

CTA200 project - Lane-Emden stars

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ABSTRACT

This project focuses on producing numerical solutions to the Lane-Emden equation using Python to simulate polytropic stars, which represents a relation between density and radius. We consider positive polytropic number values between 0 to 5 and compare them with known values for the radius where density vanishes. The resulting error between the numerical simulations and known values are mostly of order $< 10^{-4}$, except for the case of $n = 0.5$. The results from this project show that numerical integration of this equation can be used to simulate stars of varying polytropic indices using the AREPO simulation software.

1. INTRODUCTION

A polytropic star is an object that satisfies the following relation at each radius of the star,

$$P(r) = K\rho(r)^{\gamma_p} \quad (1)$$

where P is the pressure at each radius, ρ is the density, γ_p is the polytropic index, related to the polytropic number by $\gamma_p(n) = 1 + \frac{1}{n}$, and K is a constant that depends on total mass, total radius and the polytrope number given by

$$K(n) = j_n GM^{1-\frac{1}{n}} R^{\frac{3}{n}-1} \quad (2)$$

The Lane-Emden equation is

$$\frac{1}{\xi^2} \frac{d}{d\xi} \xi^2 \frac{d}{d\xi} \theta = -\theta^n \quad (3)$$

Here, $\theta = \left(\frac{\rho}{\rho_c}\right)^{\frac{1}{n}}$ with ρ_c equal to the central stellar density, and $r = \xi r_n$ with $r_n = \sqrt{\frac{(n+1)K}{4\pi G\rho_c^{1-1/n}}}$. The aim of integrating this equation is to extract a value for ξ_1 , which has the property $\xi_1 r_n = R$.

2. METHODS AND RESULTS

To solve the Lane-Emden equation, a re-parametrisation was used:

$$\begin{aligned} x_1 &= \theta \\ x_2 &= \frac{d}{d\xi} \theta \\ \Rightarrow x'_1 &= x_2 \\ x'_2 &= -\frac{2}{\xi} x_2 - x_1^n \end{aligned}$$

A function was created to output the derivatives of each re-parametrised variable and the Python module `scipy.integrate.solve_ivp` took this function as an input. Once the value of θ reached 0, the solver terminated and generated an output. Figure 1 shows the difference between the numerical and analytical solution for polytrope number $n = 1$.

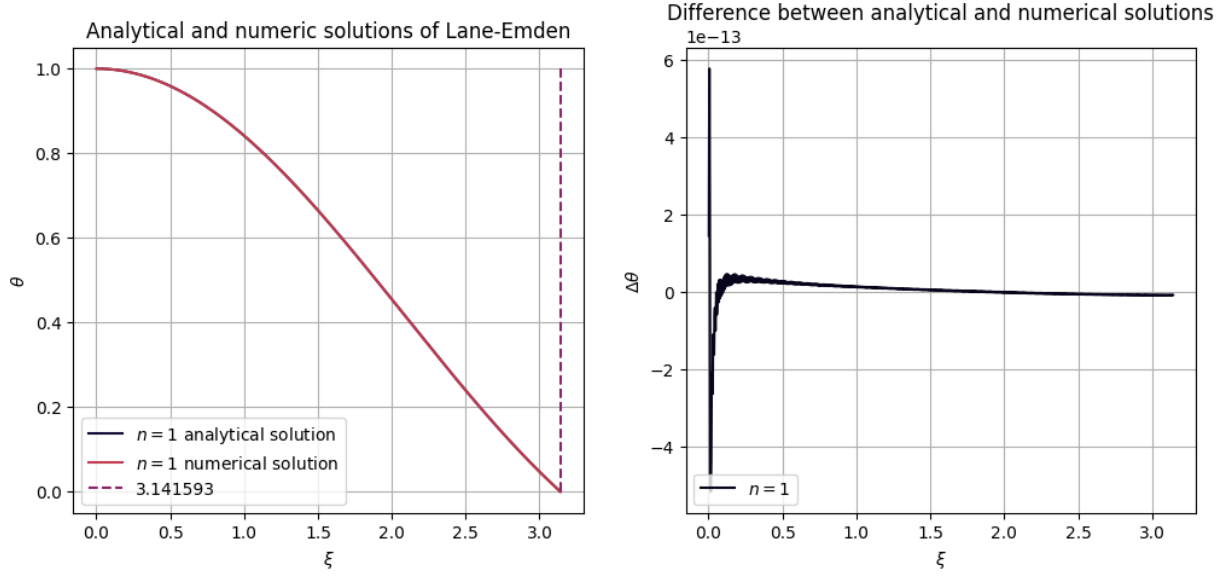


Figure 1. (left) Numerical and analytical solutions to the Lane-Emden equation for $n = 1$. (right) Residual between numerical and analytical solutions for $n = 1$

32 The solutions for other values of n were then generated, shown in Figure 2. The resulting values for ξ_1 were very
 33 close to the known values, with very little error except for the case of $n = 0.5$, as seen in Figure 3.

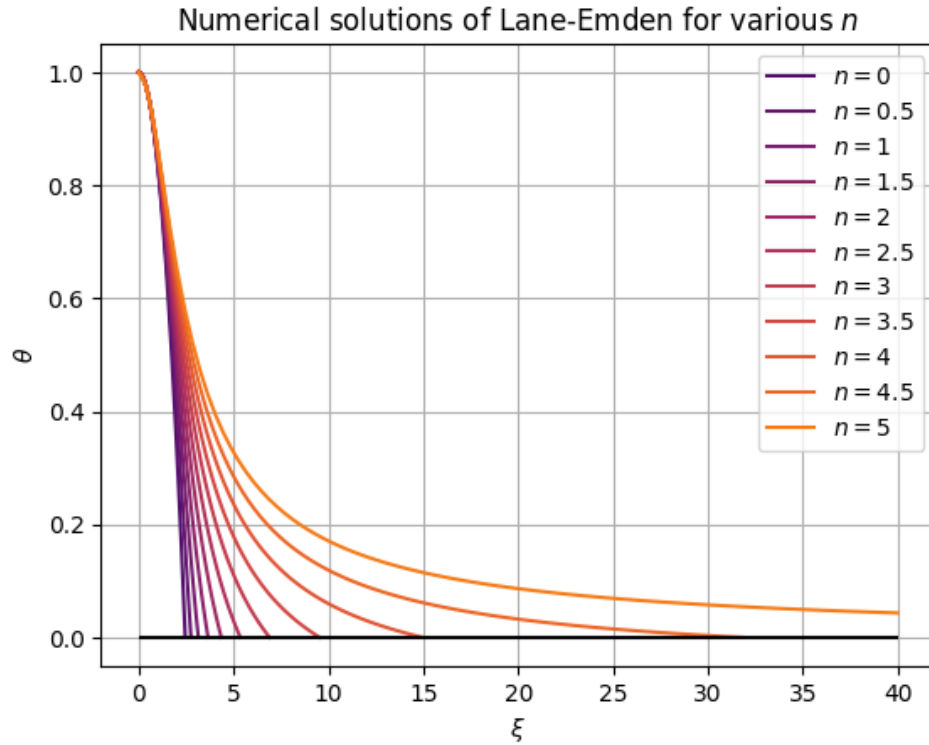


Figure 2. Solutions to the Lane-Emden equation using `scipy.integrate.solve_ivp`.

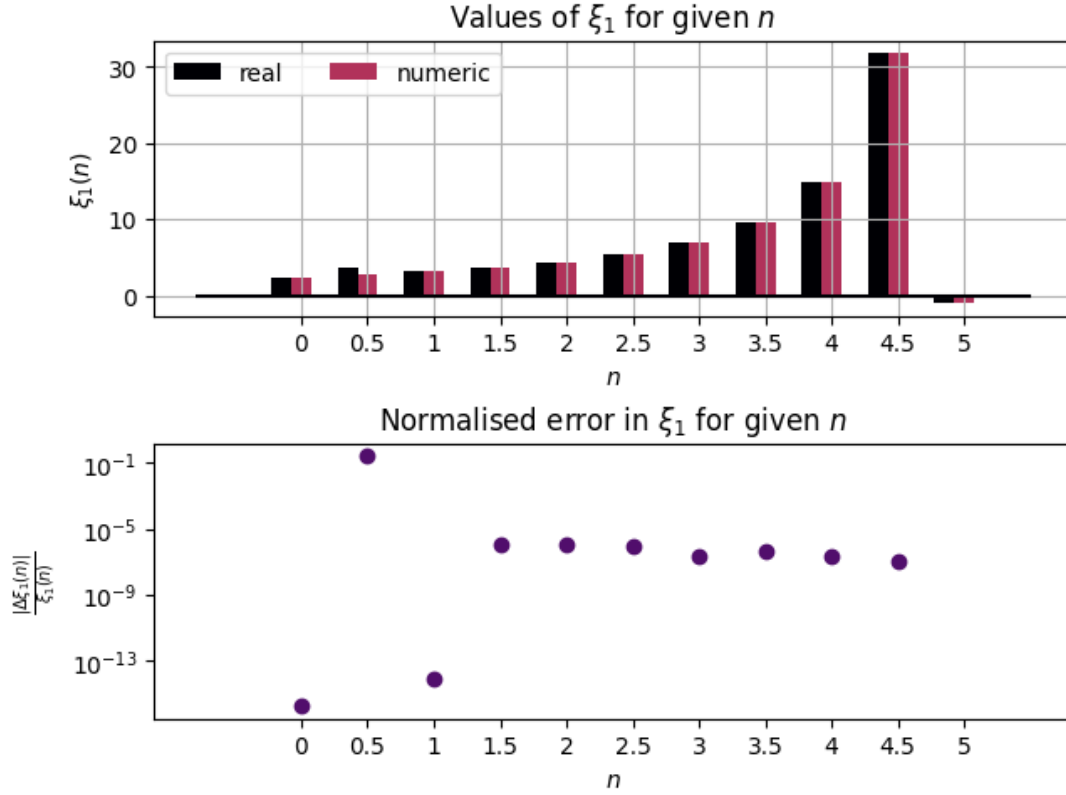


Figure 3. (top) Raw values of numerical values of ξ_1 compared to actual known values. -1 was given to $n = 5$ as the solution diverges. (bottom) Error of ξ_1 between real and numerical solutions. $n = 0.5$ gives an error of order 10^0 , whereas other values of n return errors $< 10^{-4}$

34 The argument 'max_step' in the solve_ivp function was crucial in reducing error. A smaller maximum step led to
 35 lower deviation from the true value of ξ_1 . Figure 4 shows the comparison between a solution where the maximum
 36 step was not defined and one that had a definite maximum step. Note that the errors for $n = 0, 1$ are different than
 37 that of Figure 3. This is because the residuals in Figure 4 compared the numerical solutions to values of ξ_1 with finite
 38 significant figures, which are accurate up to order 10^{-5} , whereas Figure 3 uses the analytical values of ξ_1 for $n = 0$
 39 and $n = 1$ to compare with the numerical methods.

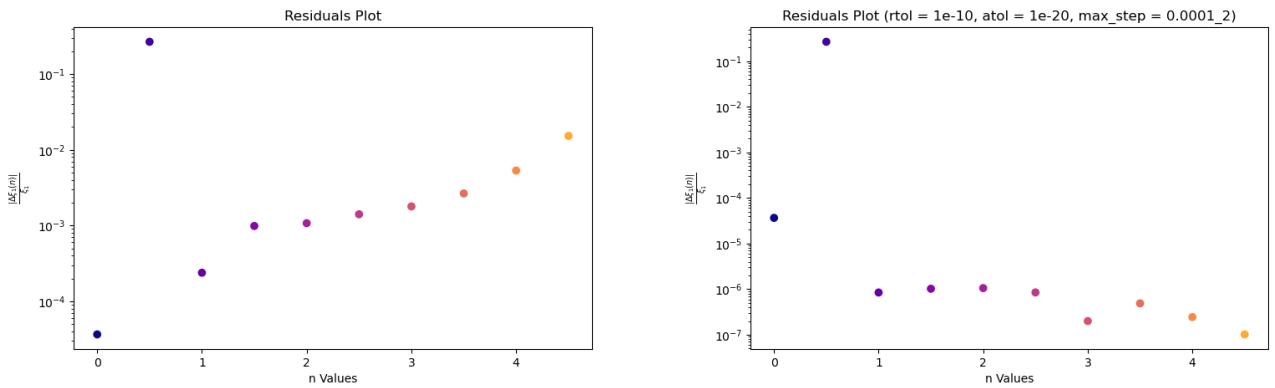


Figure 4. (left) Residuals for undefined max_step. Used default solve_ivp value. (right) Defined max_step to be equal to 10^{-4} . Errors now mostly fall below order 10^{-4}

As the error for ξ_1 is small for most n , we plotted the values of ξ_1 for continuous n , given in Figure 5. The function appears continuous and differentiable, and gives insight into the structure for objects of varying polytrope numbers.

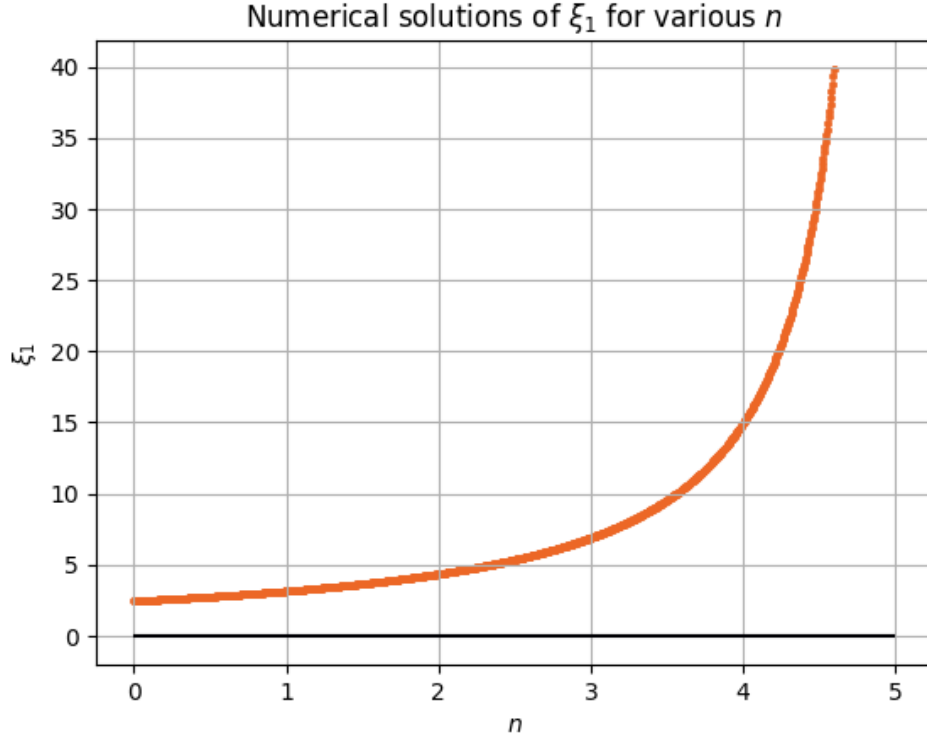


Figure 5. Values of ξ_1 for continuous n between $n = 0$ to $n = 5$

These results provide a way to obtain the density profiles and maximum radii of stars that fall under the $n = \frac{3}{2}$ to $n = 3$ polytropes, which would assist in simulating non-degenerate objects with extreme properties.

3. FUTURE PROJECTS: SIMULATIONS WITH AREPO

The numerical solutions to the Lane-Emden equation have been acquired, and so stellar simulations with varying polytropic indices can be executed through AREPO. Future projects will look into generating data sheets as input into AREPO's program, and observing the physics of the objects created by the simulation and their lives.

Software: scipy, numpy, matplotlib, astropy